

AREMU OLUWATOBILOBA

AI Engineer | Backend Engineer

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PROFESSIONAL SUMMARY

AI Engineer with a strong backend foundation, specialising in building intelligent systems powered by Large Language Models. Experienced in designing and deploying end-to-end AI pipelines, RESTful APIs using FastAPI, and vector-based retrieval systems. Passionate about applying AI to solve real-world problems across different fields and industries. Motivated, fast-learning, and driven by practical impact through software engineering.

TECHNICAL SKILLS

AI & Machine Learning: RAG Pipelines, LLM Integration, Prompt Engineering, Semantic Search, Vector Databases (ChromaDB), Sentence Transformers, Groq API, Ollama (LLaMA 3.1)

Backend Development: Python, FastAPI, Flask, REST API Design, PostgreSQL, Uvicorn

DevOps & Deployment: Docker, Hugging Face Spaces, Git, GitHub

Frontend & Interfaces: Gradio

Currently Exploring: Agentic AI Systems, Multi-agent Workflows

PROJECTS

MyScholar AI — RAG Pipeline for Academic Document Q&A | 2026

GitHub: <https://github.com/Aremuoluwatobi/rag-pipeline> | Live Demo:

<https://huggingface.co/spaces/AremuTaiwo/Rag-Pipeline>

- Built a full end-to-end Retrieval-Augmented Generation (RAG) system that enables students and researchers to query academic PDF and DOCX documents conversationally
- Designed an ingestion pipeline covering text extraction, chunking, embedding generation, and vector storage using ChromaDB and Sentence Transformers
- Integrated Groq API (llama-3.1-8b-instant) for context-aware response generation grounded strictly in document content
- Implemented exponential backoff retry logic to handle Groq API rate limits gracefully in production
- Deployed the application on Hugging Face Spaces using Docker, with documents stored and retrieved from a Hugging Face Dataset at runtime
- Built a FastAPI backend for local testing and programmatic access via Postman alongside the Gradio conversational interface
- Designed to solve real-world information retrieval problems across different fields and industries

Inventory Management API — Backend + AI Categorisation System | 2026

GitHub: <https://github.com/Aremuoluwatobi/inventory-api>

- Built a Python-based backend API for managing inventory operations with full CRUD functionality
- Designed RESTful endpoints for inventory management, admin authentication, and system workflow using Flask and PostgreSQL
- Integrated LLaMA 3.1 via Ollama to automatically categorise inventory items into predefined categories
- Applied prompt engineering to design structured instructions and constraints for accurate AI-driven classification

- Implemented JWT-based authentication and bcrypt password hashing for secure admin access

EDUCATION

Bachelor of Arts in History | University of Ilorin | 2023 – Present

INTERESTS

- AI Engineering and Large Language Model Applications
- Backend Engineering and Scalable API Systems
- Applying AI to solve real-world problems across different fields and industries
- Emerging AI — Agentic workflows, multi-agent systems, and autonomous pipelines